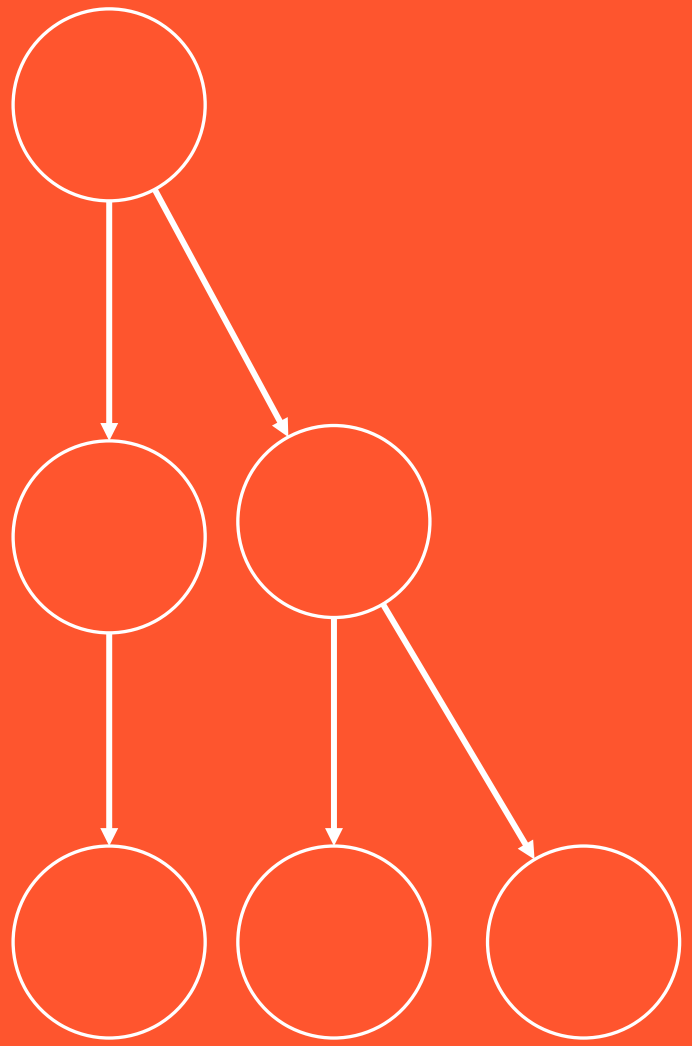


BCOG 100

Pattern Recognition

Lecture 7.2 — Neural Networks and Pattern Recognition

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Define a Chair

In your group, three minutes.

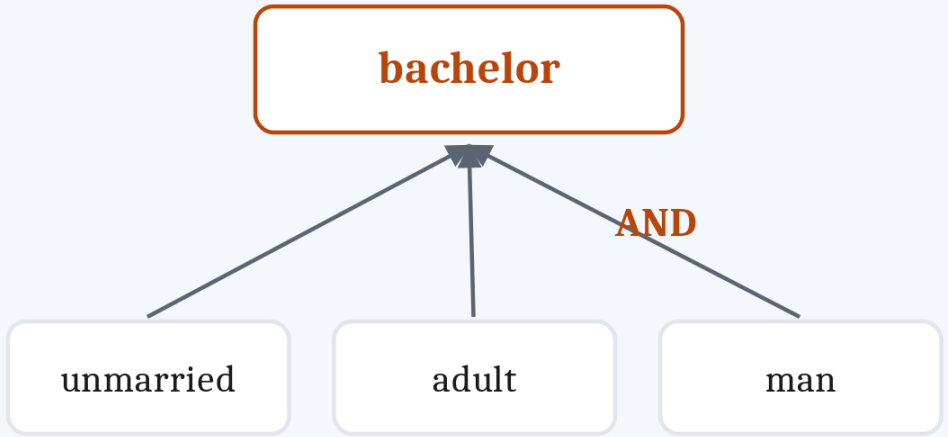
Write a definition of *chair* that every chair meets and nothing else does.

Then test it: a beanbag, a bar stool, a tree stump someone is sitting on, a chair in a dollhouse, a throne.

Definitions and Rules

The **classical view**: a category is defined by features that are individually necessary and jointly sufficient. A definition can be read as labeled-line tests combined by a rule.

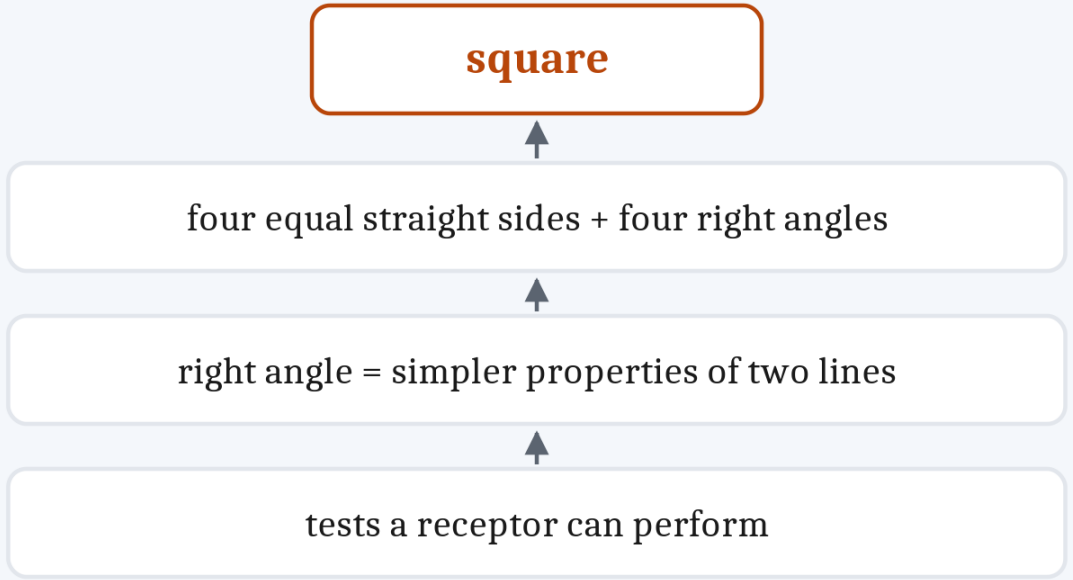
A definition: tests and a rule



fires only when every test is passed

(our translation, not the classical theorists')

Definitions can be stacked

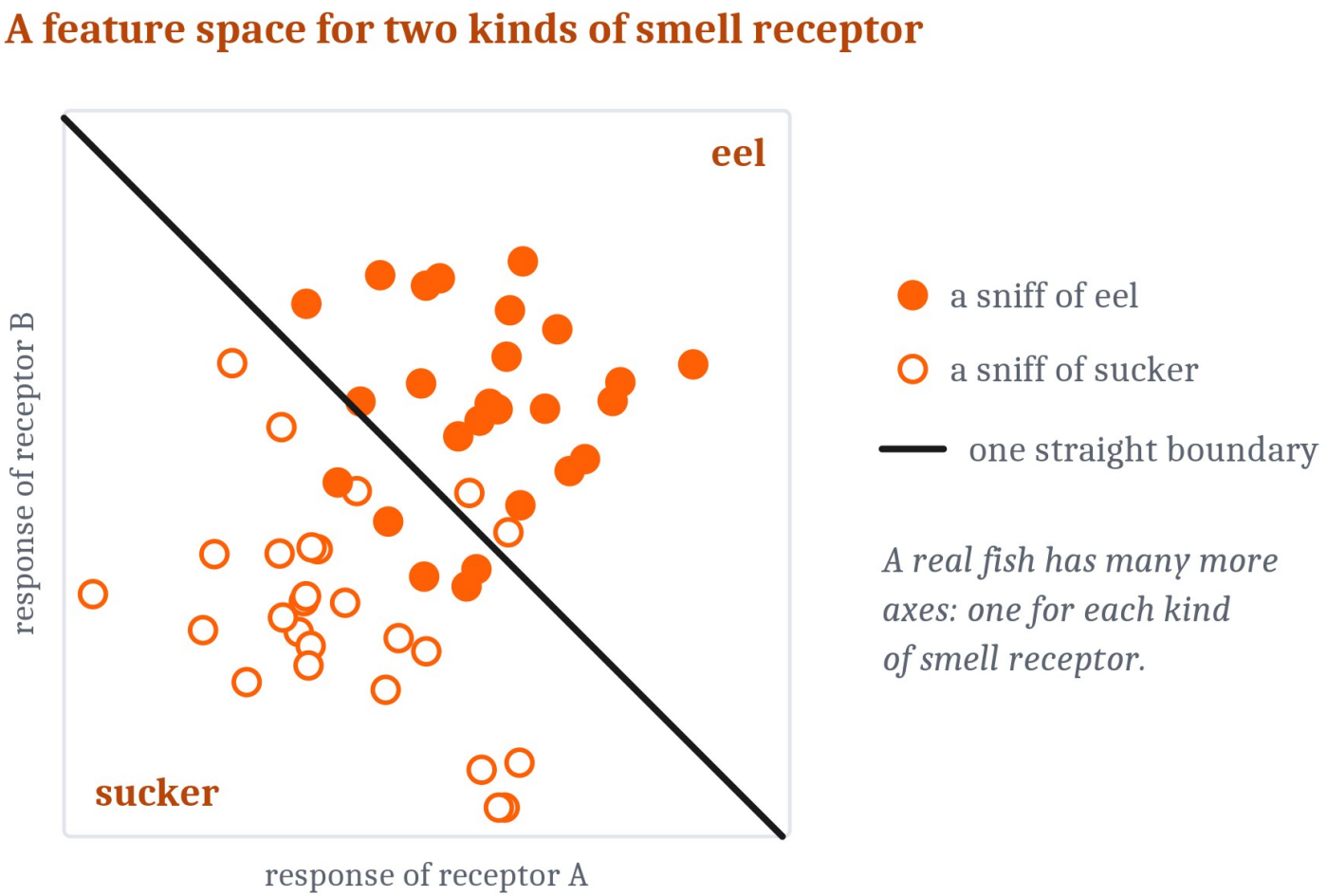


For a square the chain reaches all the way down. For bird or chair, it may not.

And people rate 3 a better odd number than 447, though every odd number belongs equally.

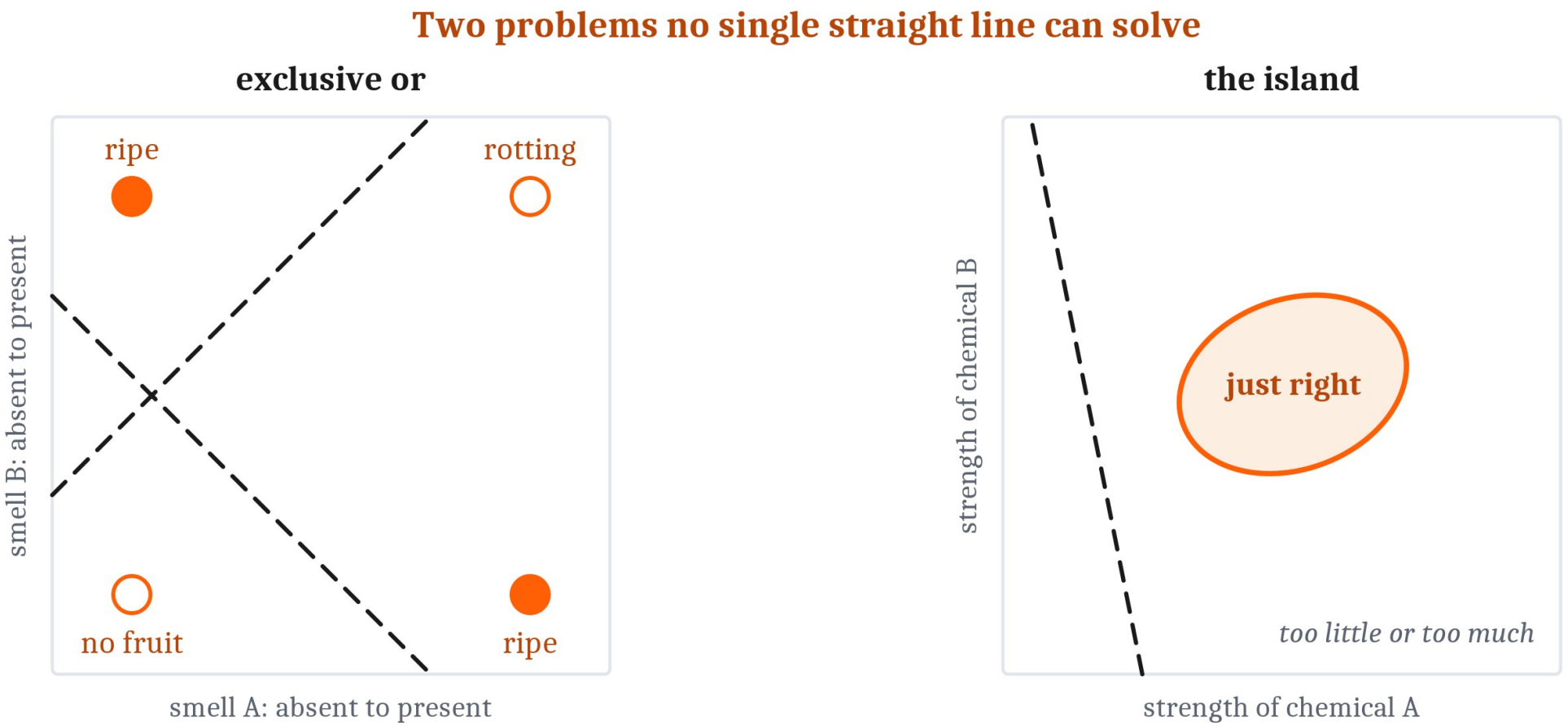
A Space of Inputs

A **feature space** has one axis per feature and one point per input. **Linear separability**: one straight line, or flat surface, can divide two categories.



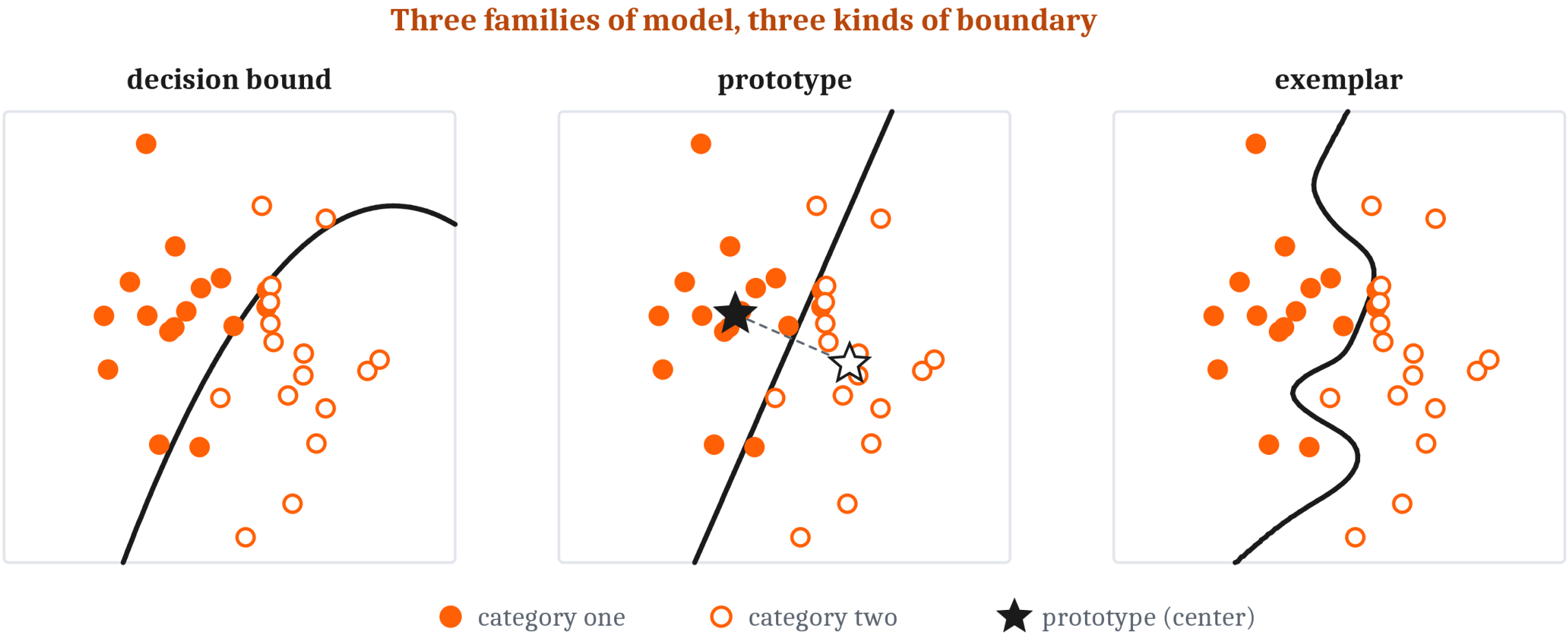
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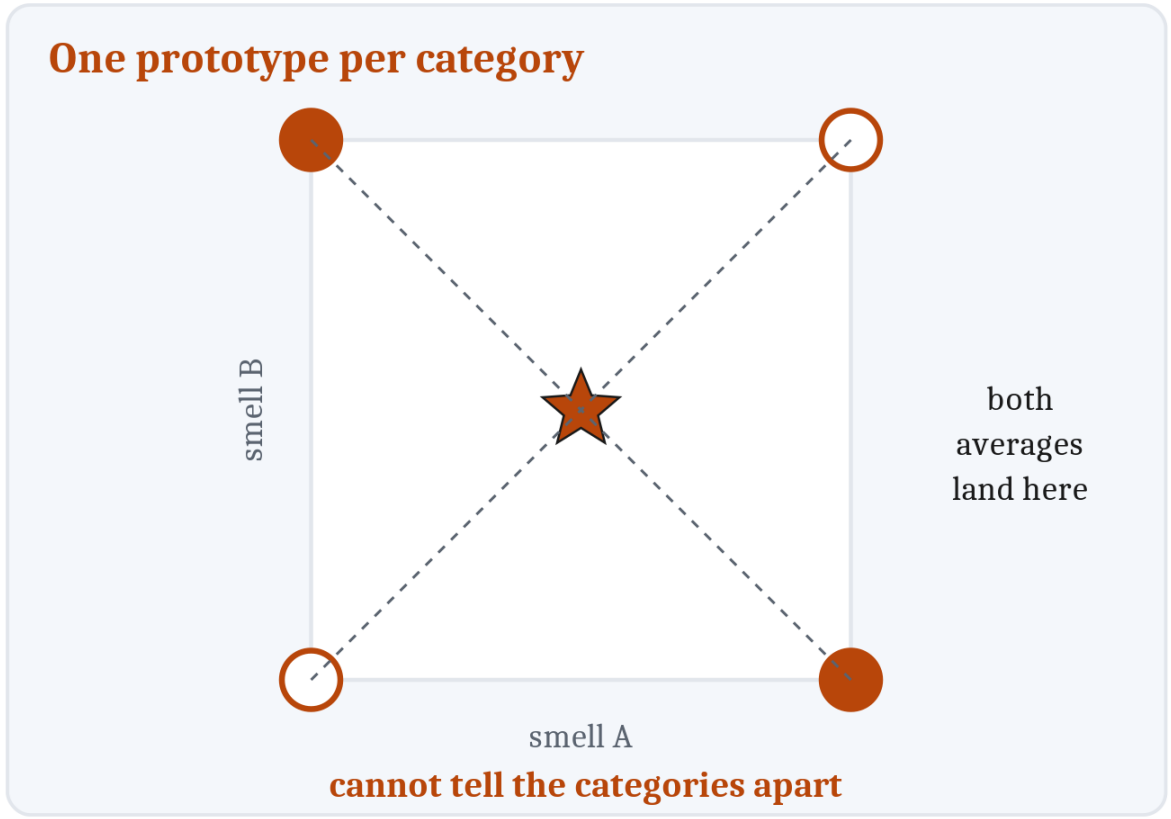
Three Families of Algorithm

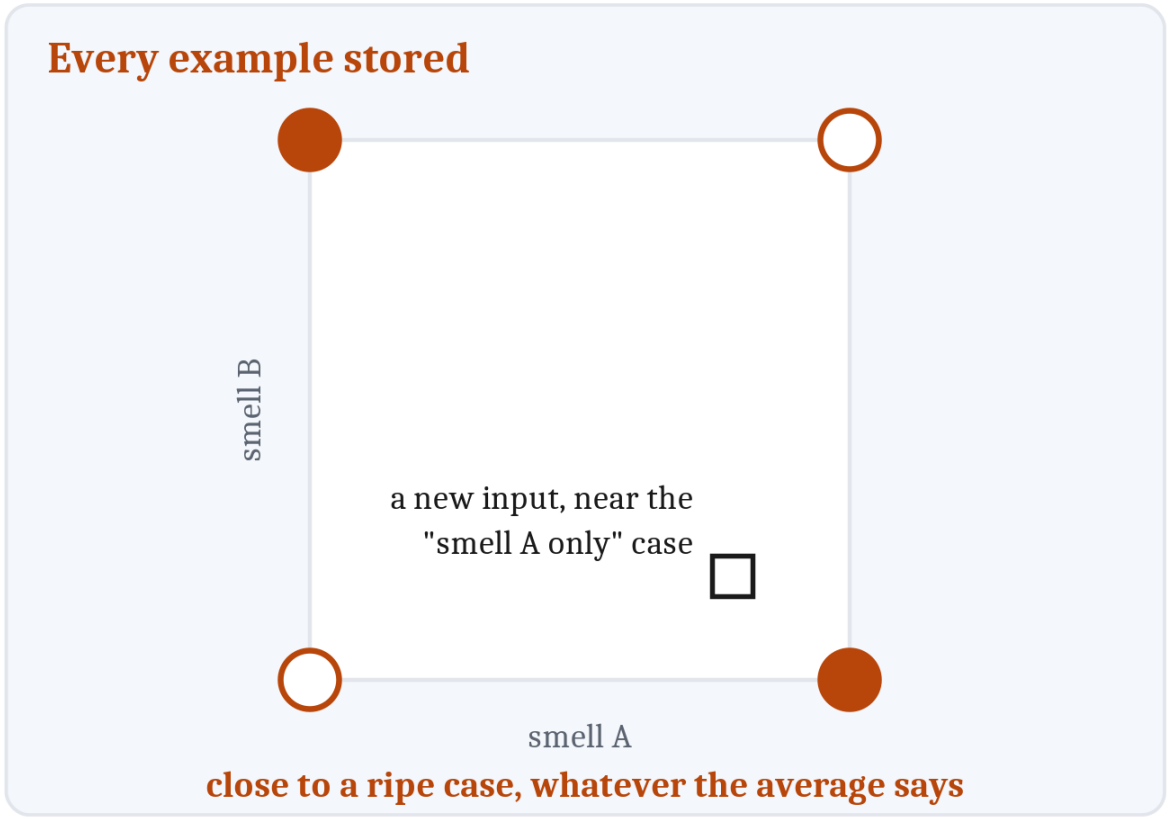
A **decision-bound model** stores the boundary. A **prototype** model stores each category's center; an **exemplar** model stores every member. Both decide by **similarity**.



Prototypes or Exemplars?

One prototype per category can only sort linearly separable categories. Exemplars keep the cases. Different in principle, hard to tell apart in practice.



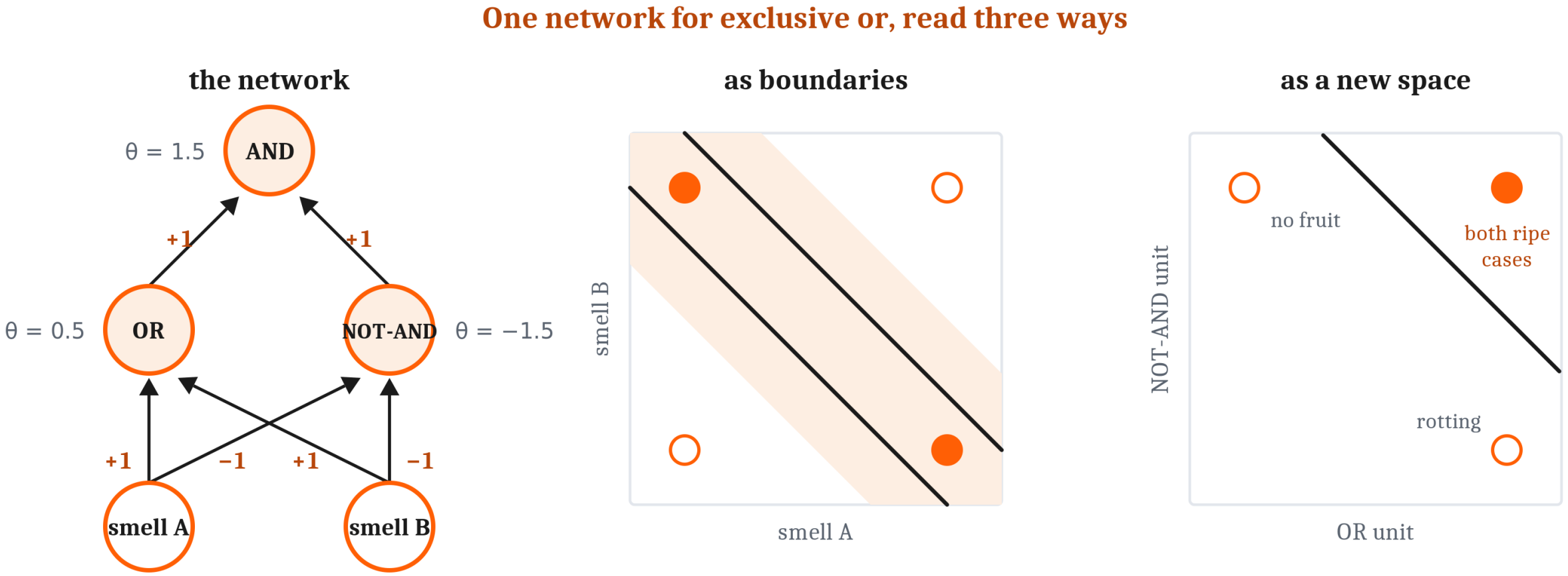


Filled: ripe fruit. Hollow: no fruit, or rotting. Star: the prototype of each category. Square: a new input.

Different in principle. Hard to tell apart in practice. No consensus.

A Layer Between Input and Output

A **hidden layer** re-represents the input so that the last step is easy. The second hidden unit means “not rotting”, which no receptor supplies.



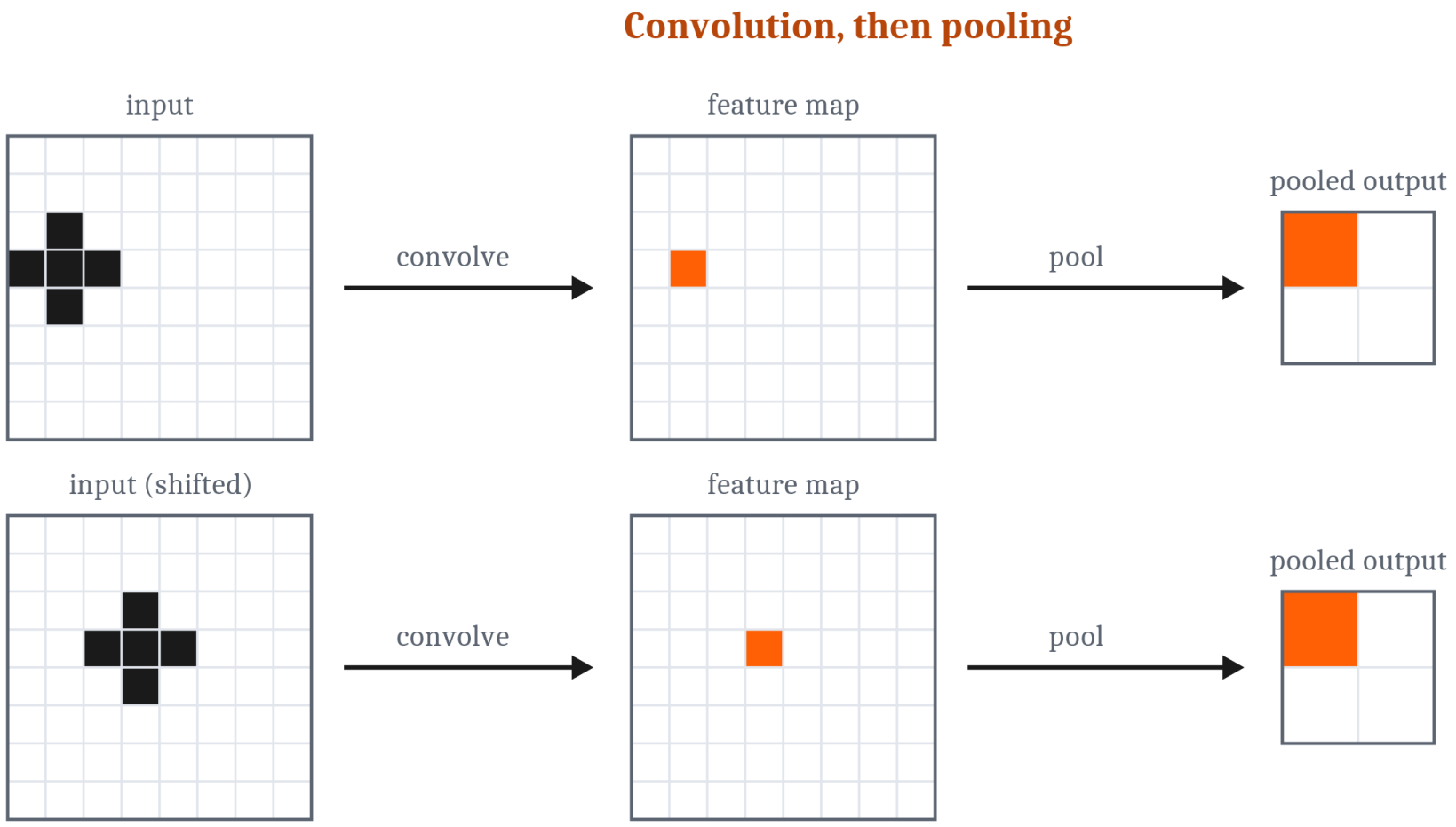
Three Ways to Set the Hidden Weights

Structural credit assignment: when the output is wrong, which connections inside were responsible?

Answer	What weight changes depend on	Which weights change	What the hidden layer comes to represent	A named case
Backpropagation (error-driven)	An error against a target, from a teacher or the world's next input	Every weight	Whatever combinations reduce the error	The digit-reading network; Elman's next-word networks
Fixed random wiring, trained output	For the output connections only, an error or a reward	Only hidden-to-output	Whatever combinations the random wiring produced	The perceptron; the insect mushroom body, as proposed
Competitive learning	Nothing outside the network	Only the winner's weights	Clusters of inputs that overlap, useful or not	Topographic maps; the neocognitron

Stacking Layers for Invariance

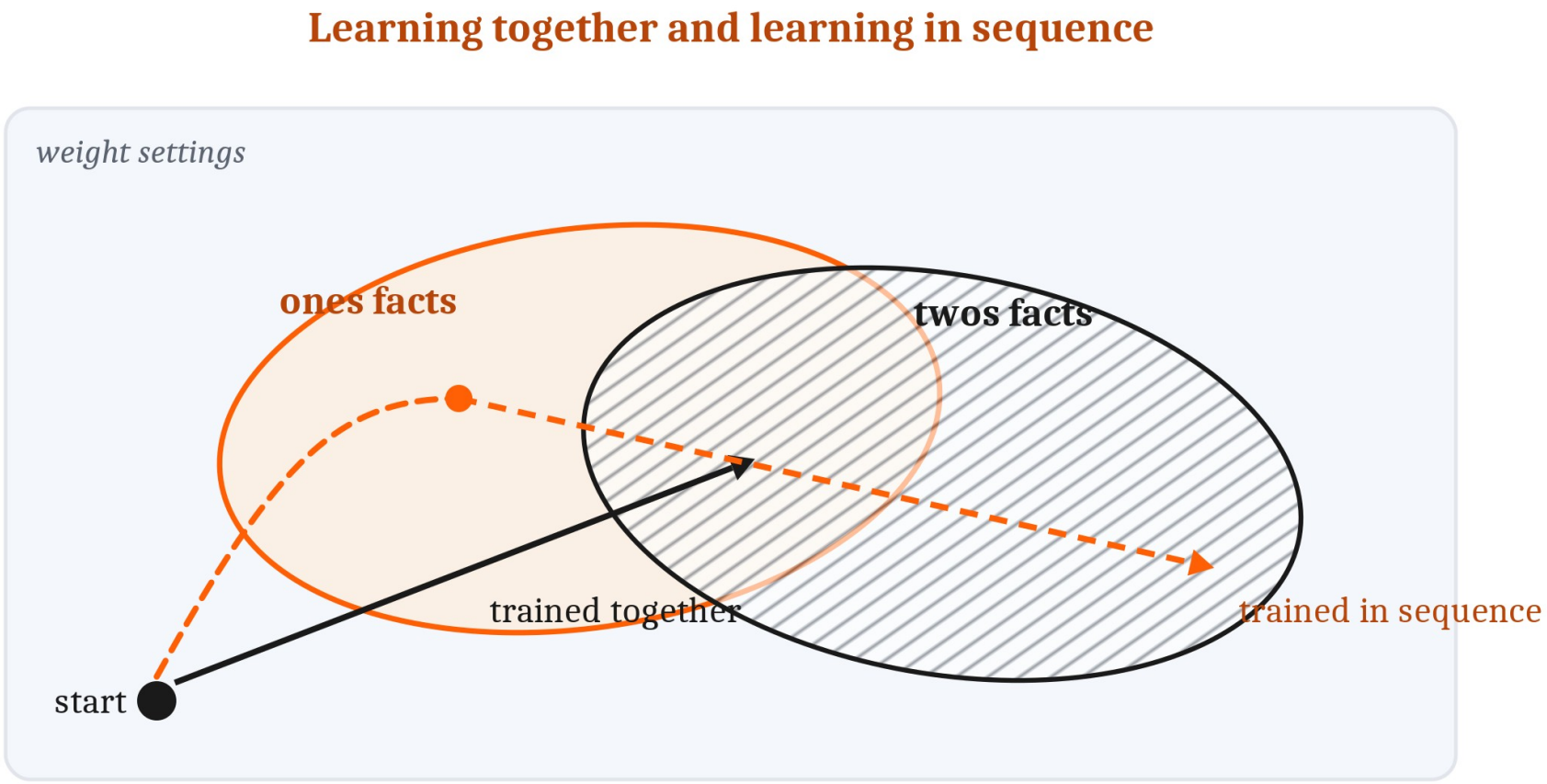
Convolution copies one detector to every position, so its map shifts with the input. **Pooling** makes a small shift stop mattering. Both are an **inductive bias**.



The detection moves with the input; the pooled output does not change.

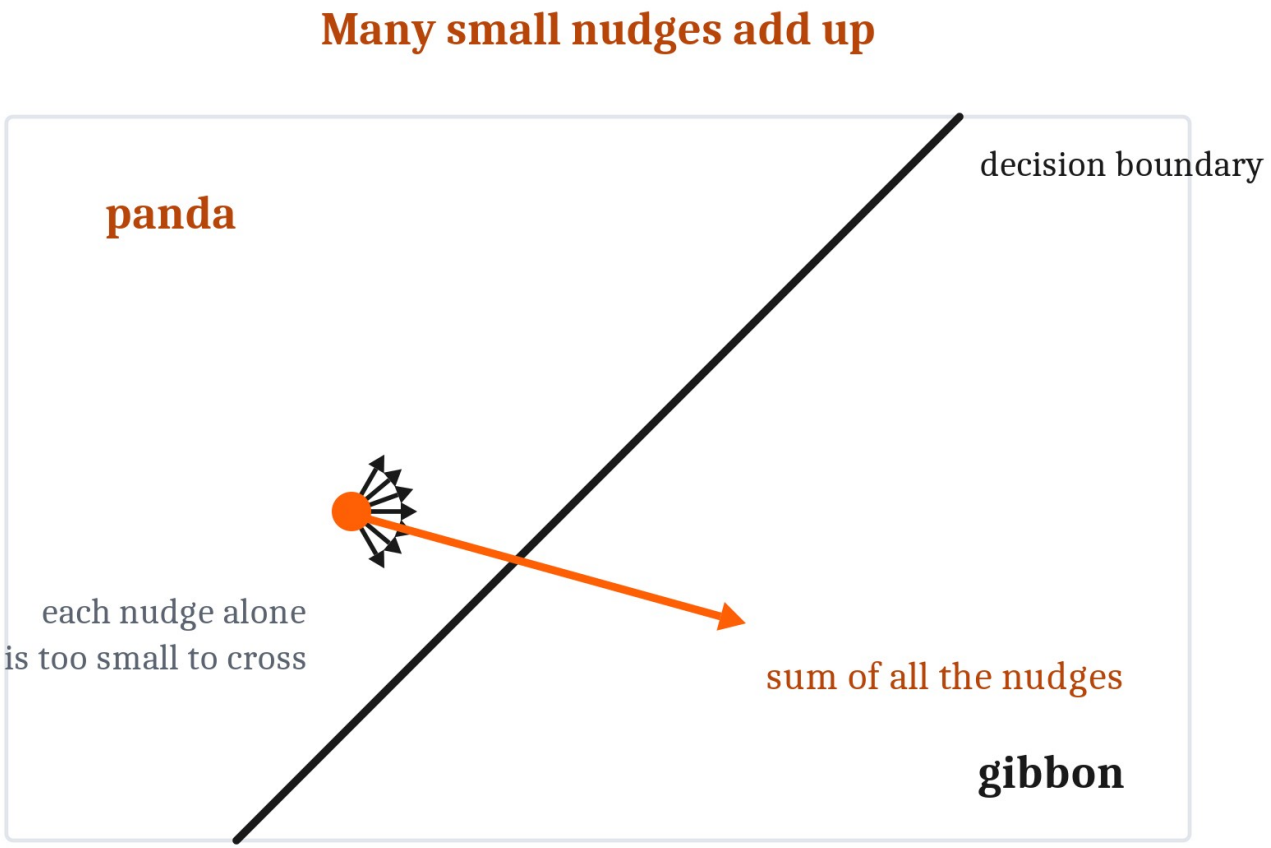
What Shared Weights Give, and Take Away

Catastrophic interference: new learning wipes out old. Generalization and interference are the same effect, seen from two sides; learning in sequence is the trigger.



Fooled by Small Changes, Lost Far from Home

Adversarial examples, **texture bias**, **overfitting**, and weak **out-of-distribution generalization**: the characteristic failures, and some of their reasons.



Gaps in Recognition, or Something Else Missing?

Not every failure is the same kind of failure. **Systematicity**: combining known parts in new ways.

Failure	What kind of failure it is	Where we take it up
Adversarial examples	A gap in pattern recognition	This module
Texture bias	A gap in pattern recognition, produced by the training data	This module
Overfitting	A gap in pattern recognition, reduced by inductive bias	This module
Failure out of distribution	A gap in pattern recognition	This module
Catastrophic interference	A gap in the learning procedure, under learning in sequence	This module; how brains avoid it, Module 10
Knowing what an object is like, beyond recognizing it	A mark of a later capacity	Modules 8 and 12
Reasoning about causes	A mark of a later capacity	Module 12
Systematicity	A mark of a later capacity	Modules 13 and 14

What Has to Be Built In?

Not “innate or learned?” but *what exactly* is built in. Four candidates, and two reasons not to build in everything.

1. The input space

which receptors an animal has:
what differences it can register

almost no one doubts it

2. An architectural bias

convolution; a large random
expansion layer

few doubt some of these

3. The learning rule

what a weight change depends on:
flavor with illness, not shock

what it can extract: disputed

4. Specific content

a detector wired for one predator;
newborns following a face

how much is supplied: disputed

Why not build everything in?

Size

10 features: 45 pairs, 120 triples, more than
a thousand combinations of all sizes. A fish has
many dozens of kinds of smell receptor.

Change

A built-in detector fits the ancestors' world.
A new predator or a new food: learning can follow,
and a built-in detector cannot.
Building in too much overfits the lineage's past.

Which domain gets which answer is an empirical question.

Sort the Failure

In your group, four minutes. Using the table on gaps and marks, sort each case, and say what evidence would change your answer.

1. A few small stickers on a stop sign, and a classifier reads it as a speed-limit sign.
2. A skin-lesion classifier gives higher cancer scores when the photo shows the surgical-marker ink dermatologists use to mark a mole.
3. A model names every object on a kitchen shelf, but cannot say what would happen if the shelf gave way.

Image Sources (1 of 2)

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3	A definition as tests and a rule, and stacked	fig-definition-tree.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
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6	Why one prototype cannot learn exclusive or	fig-one-prototype.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
7	One network for exclusive or, read three ways	fig-xor-three-ways.png	CC BY 4.0	Jon Willits — drawn for the Module 7 reading · own work
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